

BINAY- Lighting Up The Future

Binay LED Aviation Obstruction Lights



**BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon**

As per the International Civil Aviation Organisation (ICAO) requirements for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per the International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



**THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT**



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gold nanoparticles into organic photovoltaic solar cells helps enhance the cells' power conversion efficiency. By employing the interconnecting gold nanoparticles layer, they were able to enhance power conversion by as much as 20 per cent.

The gold nanoparticles create a strong electromagnetic field inside the thin organic photovoltaic layers by a plasmonic effect, which concentrates light so that much more of it can be absorbed by the subcells.

Human brain like chips

IBM is taking a key step in training a computer to behave like a human brain, both technologically and physiologically. The company has built two prototype chips that process data more like humans digest information than the chips that now power PCs and supercomputers and are an evidence of the growing importance of multi-tasking computers.

A six-year-long project involving a hundred researchers and around \$41 million funding from the US government's Defense Advanced Research Projects Agency (DARPA) resulted into the production of these chips. It may take another decade for the chips to be out of the lab and find their place inside products.

Solar cells to power soldiers

Infantry soldiers are increasingly being equipped with electronic devices to enhance their close-combat tactical awareness. Currently, soldiers are dependent on electrical power provided by a conventional battery to power these devices. Each battery has a different endurance and reliability level and each rechargeable type requires its own kit, compounding the bulk and weight that needs to be carried.

The Australian National University (ANU) for Sustainable Energy Systems has developed some tough, lightweight solar panels that will now allow soldiers to generate power in the field and reduce the need for batteries for their electronic devices. These will keep electronic devices operational throughout the duration of missions.

Professor Andrew Blakers, project's chief Investigator, said that SLIVER solar cell technology formulated by ANU, forms the groundwork of these wearable solar panels. SLIVER cells have enabled the construction of efficient, rugged, flexible and light-weight portable modules that convert light directly into electricity under a wide range of environmental conditions. These cells are as thin as a sheet of paper or a human hair. This means they are flexible, lightweight and allow high power-to-weight ratios to be achieved. The cells are also bifacial, allowing modules to be constructed such that light can be absorbed from both faces.

Nano-structured glass for imaging and recording

Researchers at the University of Southampton have developed nano-structured glass optical elements which have applications in optical manipulation and will reduce the cost